

EXECUTIVE SUMMARY

SEL-002: Bandon Public Transport Accessibility

9,863 AGGREGATE POPULATION Across 34 Census Small Areas in Bandon BUA	64.7% WALK ACCESS DEFICIT 22 of 34 areas require >10 min walk to transit	32.4% OUTSIDE CATCHMENT 11 areas sit completely outside 15-min walk zone
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Core Findings & Value Proposition

This independent geospatial analysis establishes a spatial screening framework across 34 Census Small Areas intersecting the Bandon Built-Up Area (BUA study population: 8,196; aggregate Small Area population: 9,863). The research decouples transport network performance from population demographic need to identify systemic deficits and targeted investment opportunities.

- Centralised Network Bias:** 64.7% of evaluated Small Areas (22 of 34) require network walking times exceeding 10 minutes to reach principal transit stops, with 32.4% (11 Small Areas) falling completely outside the 15-minute network walking catchment (>1,250 m).
- Strategic Hotspot Isolation:** By combining an Access Deficit measure (40% weight) with a normalized Population Vulnerability Index (60% weight—encompassing older age, health/disability, car availability, and economic status), the model isolated 4 critical hotspot Small Areas. These areas, including Gully, Cloughmacsimon, and Kilbrogan, record Hotspot Scores ranging from 0.856 to 0.947, reflecting the spatial overlap of substantial accessibility deficits (Access Deficit > 0.75) and high relative population vulnerability (VUL_norm > 0.75).

Hotspot Evaluation Model Architecture

COMPOSITE HOTSPOT SCORE (100%)	
ACCESS DEFICIT (40% Weight)	POPULATION VULNERABILITY (60% Weight)
Pedestrian Walkability <i>Sub-weight: 50% Weight</i> GTFS Scheduled Peak Frequency <i>Sub-weight: 50% Weight</i>	Elderly Population Rate <i>Sub-weight: 30% Weight</i> Health & Disability Ratio <i>Sub-weight: 30% Weight</i> No-Car Household Ratio <i>Sub-weight: 20% Weight</i> Economic Vulnerability <i>Sub-weight: 20% Weight</i>

Critical Priority Hotspot Small Areas

Small Area / Location	Access Deficit	Vulnerability Index	Composite Score
Gully Peripheral Zone	0.892 (Severe)	0.884 (High)	0.947
Cloughmacsimon Area	0.810 (High)	0.865 (High)	0.895
Kilbrogan Outer Sector	0.785 (High)	0.820 (High)	0.868
Bandon West Corridor	0.760 (High)	0.790 (Moderate)	0.856

Capital Allocation & Operational Recommendations

Targeted Active Travel Infrastructure

Prioritise footpath connectivity, pedestrian crossings, and gradient improvements along arterial routes connecting outer residential areas (Gully, Cloughmacsimon, Kilbrogan) to central transit nodes.

Service Frequency Optimisation

Re-evaluate GTFS scheduled peak-period headways where pedestrian network access is high but service frequency remains low.

Demand-Responsive Transit (DRT)

Conduct feasibility assessments for micro-transit or DRT solutions in outer peripheral zones situated beyond the 15-minute walking catchment, rather than extending costly fixed-route infrastructure.